

Execution of Modal Triplet Theory I: Gauge, HYM, Threshold, and Axion Status after Profile-Standard SM Closure

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Version 3

Abstract

We report the current MTT gauge, Hermitian Yang–Mills (HYM), threshold, and axion execution. The selected gauge profile is transported by SMDR v1.3 to $Q = M_t$ and enters an eight-row positive-definite precision object; embedded renormalized-Standard-Model equivalence closes at the adopted one-shared-physical-primitive/profile standard. Native bundle automorphisms give the faithful gauge group $(SU(3) \times SU(2) \times U(1))/\mathbb{Z}_6$, while the selected finite gauge/ghost spectrum source closes 10/10 with zero continuous parameters. The normalized finite rows are universal, so they do not derive the measured relative gauge couplings. On the selected $q79/F/m1$ rank-two bundle, the literal Cech–HYM witness closes through finite cocycle and continuum-tail certificates. The internal $K_{\text{threshold}}$ response ledger also closes, but physical threshold and mass-scheme values remain profile data. Strong CP remains open because the complete E_6 color anomaly cancels and no selected flux/threshold axion-current map supplies the required nonzero low-energy effective anomaly.

Version 3 Revision Note

Supersedes *Execution of Modal Triplet Theory I: Gauge, Axion, and Threshold Sectors*, version 2.

Reason The former execution relied on a fitted few-TeV Calabi–Yau benchmark, obsolete moduli and threshold values, and axion normalizations not selected by the current q79 source. It also preceded the native gauge-group and 10/10 finite gauge-spectrum execution.

Resolution Version 3 replaces that benchmark with the SMDR precision profile, the certified q79/F/m1 Cech–HYM witness, the closed internal threshold ledger, the native finite gauge results, and an explicit strong-CP source audit.

Retained result Gauge/HYM and threshold structures support embedded renormalized-SM equivalence at the adopted profile standard.

Remaining boundary Physical visible–hidden HYM endpoints, continuum gauge/action response, strict threshold values, and a selected nonzero effective color-anomaly map remain stronger upgrades.

1 Status standard and authority

This paper reports two distinct closure standards:

1. **Adopted profile standard.** MTT embeds the renormalized Standard Model on the selected branch, with one shared physical primitive and measured common-scheme profile rows. The final audit closes 12/12 obligations.

2. **Strict source standard.** Every numerical value must be emitted from selected MTT source geometry without empirical profile rows. This stronger zero-primitive/no-knob standard remains open.

The distinction is essential: closure at the first standard must not be described as closure at the second.

The old Tier-3/Tier-4 calculation used a gauge crossing near 5 TeV to fit Kahler ratios, thresholds, and axion data. That crossing is withdrawn. None of its downstream numerical geometry is retained as current authority.

2 Selected gauge and precision execution

The selected common-scheme point is

$$Q = M_t = 172.5590883453979 \text{ GeV}.$$

The gauge rows are

$$g_Y = 0.3585945042 \pm 0.0000307251, \quad (2.1)$$

$$g_2 = 0.6475986708 \pm 0.0000287665, \quad (2.2)$$

$$g_3 = 1.163427409 \pm 0.004036156, \quad (2.3)$$

with $g_1 = \sqrt{5/3} g_Y$ when GUT normalization is required. These are measured-profile coordinates transported by the selected multi-loop Standard Model pipeline; they are not first-principles MTT predictions.

The full selected precision map transports 15 common source coordinates into eight $\overline{\text{MS}}$ output rows. Its covariance is positive definite, contains all 36 symmetric entries, and contains all 15 BCT–WZH cross-block entries. The declared reproducible baseline is the diagonal measured-input profile because no public official joint likelihood for all 15 coordinates was identified.

3 Native gauge carrier and finite spectrum

The rank-one, rank-two, and rank-three native bundle tensors have automorphism groups $U(1)$, $USp(2) = SU(2)$, and $SU(3)$ respectively. Their action on the selected chiral carrier has diagonal kernel $\mathbb{Z}_6$, giving the faithful low-energy group

$$G_{\text{SM}} = \frac{SU(3) \times SU(2) \times U(1)}{\mathbb{Z}_6}.$$

The associated 48-state three-family representation contains the Q, u^c, d^c, L, e^c, N^c rows and passes the cubic, mixed, gravitational, and global Witten anomaly checks. This is an exact finite-carrier result, subject to the selected low-energy branch; it is not a derivation of the measured coupling strengths.

The gauge-fixed fluctuation complex also derives the standard one-loop coefficient vector

$$b = \left(\frac{41}{10}, -\frac{19}{6}, -7 \right)$$

from the selected chiral, Higgs, gauge, and ghost representation content. At the finite spectral level, the nonabelian gauge/ghost rows are

	0	$4\pi^2/9$	$8\pi^2/9$
$SU(2)$	3	12	12
$SU(3)$	8	32	32

where the entries are multiplicities. Together with the abelian and matter rows, the finite spectrum-source inventory closes 10/10 exactly and adds no continuous parameter.

This execution comes with a no-go result that fixes its interpretation. After normalization per rank or adjoint lane, every nontrivially weighted finite row has the same shape. Hence the finite spectrum alone cannot produce the observed relative gauge kinetic coefficients. A scalar proper-time measure also leaves their ratios invariant. The physical exit is therefore a nonuniversal continuum overlap/action response on the selected q79 visible–hidden geometry, not another finite-spectrum fit.

4 HYM and bundle execution

The selected literal bundle result is stronger than the legacy string-lift ansatz. On the selected q79/ $F/m1$ rank-two bundle:

- the finite Cech witness contains 81 table entries and all 729 cocycle triples;
- the projected HYM solution has residual 8.21×10^{-13} , coercivity margin 26.02, and error indicator 3.15×10^{-14} ; and
- the weighted-theta Fourier-tail/Wiener certificate has $Z = 0.38508$ and $Y + Zr = 0.00932703 < r = 0.01$.

Together these close the selected rank-two literal Cech–HYM witness families 2/2, including continuum existence and local uniqueness in the certified ball. They do not prove uniqueness over all HYM branches or rank-three continuum bundle transfer. The finite rank-three color spectrum is closed, but that result is not yet an evaluation of a nonzero-Chern physical rank-three HYM connection.

This q79/Fu–Yau bundle result is not an identification with the auxiliary $L(3, 1) \times \text{Nil}_3$ overlap model used in the Theta papers. The shared circle is common $U(1)$ phase/holonomy data in the q79 carrier and is not an extra Cartesian factor. A same-source world-in-world/strain-to-q79 intertwining theorem remains a separate globalization target. The physical visible and hidden holomorphic bundles in one Hull–Strominger/HYM chamber, and the continuum gauge Hessian evaluated on them, remain open source objects.

5 Threshold execution

Two threshold layers must be kept separate.

5.1 Internal response layer

The selected internal $K_{\text{threshold}}$ ledger closes 10/10. The direct $K_{\text{threshold}} \cdot \Omega_H \cdot \lambda$ row is locked, and the charged K -to- Ω construction is available through the selected source rule

$$\Omega_i^{\text{src}} = D_{\text{fin}}[\text{class}(i)] K_{\text{threshold}} \cdot \Omega_i e^{-2\pi n_i}.$$

This is a closure statement about the finite internal response machinery.

5.2 Physical matching layer

Seven physical threshold rows and three mass-scheme rows are closed only at the admitted-external/profile tier. Their strict MTT-emitted value count remains zero. The selected multi-loop transport and covariance therefore make the profile execution reproducible, but they do not convert measured matching rows into source-derived constants.

The legacy one-loop bulk logarithm, exceptional-cycle vector, and minimum-norm threshold fit are withdrawn as numerical results. They may be retained only as historical ansatz templates for a future source-selected compactification.

6 Axion and strong-CP status

The current corpus supports a conditional Peccei–Quinn mechanism and selected axion decay-constant ratios. It does not select the absolute axion normalization or solve strong CP.

For the complete E_6 Q_ψ current, the exact color-anomaly audit gives

$$\mathcal{A}_{\text{matter}} = +12, \quad \mathcal{A}_{\text{complete-27 exotics}} = -12, \quad \mathcal{A}_{\text{total}} = 0.$$

The matter-only/singlet diagnostic $N_{\text{DW}} = 3$ is not the anomaly of the complete spectrum. A strong-CP theorem therefore requires a selected flux/threshold decoupling and axion-current anomaly-matching map, followed by quality and electric-dipole-moment control. That map remains open.

7 How the layers fit together

The finite carrier fixes which gauge transformations and representation channels exist. The finite gauge/ghost spectra then determine exact internal mode shapes and multiplicities. The rank-two HYM theorem proves that one selected continuum connection exists and is locally controlled. Finally, the SMDR profile transports measured physical coordinates in a declared renormalization scheme. These layers are compatible, but they answer different questions.

In particular, exact multiplicities do not fix a dimensionful normalization, and one HYM connection representative does not yet provide all physical visible–hidden overlap coefficients. The profile execution demonstrates that the architecture can carry the renormalized Standard Model. A strict prediction would additionally require the same selected continuum action to emit the relative gauge metrics, electroweak primitive, and matching rows without using those observables in source selection.

8 Retired Calabi–Yau benchmark

The former values $\zeta_2/\zeta_1 \simeq 0.560$, $\zeta_3/\zeta_1 \simeq 0.229$, their derived Kahler ratios, the fitted internal volume, exceptional-cycle coefficients, and corresponding axion decay constants all depended on the withdrawn gauge profile. They are not updated by substituting new ratios into the old formulas because topology, stability, warping, normalization, and degeneracy were not independently selected.

A future geometric execution must publish:

1. the compactification topology, intersection form, bundle, flux and stability chamber;
2. all priors, constraints, discrete branches, flat directions and degeneracies;
3. the absolute-unit bridge and threshold scheme; and
4. a held-out comparison not used to select the geometry.

9 Conclusion

Execution I now has a clean split. Gauge/precision profile execution, the native faithful gauge group, 10/10 finite gauge/ghost spectra, selected rank-two literal Cech–HYM witness, and

internal $K_{\text{threshold}}$ machinery are closed at their declared standards. Physical coupling, threshold, and mass-scheme values remain admitted profile rows, and the old Calabi–Yau numerical lift is retired. Axion ratios and a conditional PQ lane survive, while absolute normalization and strong-CP selection remain open.

The result supports embedded renormalized-SM equivalence at the adopted profile standard. It does not establish a zero-knob compactification, derive the measured gauge rows, identify the complete physical visible–hidden HYM background, or solve strong CP.

Computational Evidence and Reproducibility

This execution paper directly reports the gauge, HYM, threshold, finite-matrix, anomaly, and precision packets listed below. Flavor and neutral-sector rows are included only to locate this paper inside the common profile-standard closure. The strict no-knob upgrade remains open.

The referenced rows are frozen to the curated results repository state identified below. Hashes are grouped in eight-character blocks for line breaking.

Repository: <https://github.com/PeterNero/mtt-results-repro>
Commit: 31247ebb 5c22f3fb b5443024 365433c6 ee0bff4a
Manifest: release/result_manifest.json
Manifest SHA-256:
fb399689 60b00584 631dbf53 1a708e18
ef928d6b 6d935119 c185d7f6 32b1e7cd

Tier labels are quoted verbatim from that manifest. A row used directly supports only the specific computational statement identified above; a corpus-state cross-check does not prove this paper’s local theorems; and an open row is evidence of an unresolved obligation, never of closure.

Rows used directly in this paper.

- A01/current_global_lock (*profile replay*).
Current non-looping global status and source-certificate map.
- A01/direct_k_higgs_row (*derived exact*).
Promoted direct $K_{\text{threshold}}.\Omega_{\text{H}}.\lambda$ row.
- A22/e6_qpsi_qcd_anomaly (*derived exact*).
E6 Qpsi matter/exotic QCD anomaly cancellation audit.
- A04/final_12_of_12_audit (*profile replay*).
Twelve-obligation embedded renormalized-SM equivalence audit.
- A19/hym_wiener_contraction (*numeric certified*).
Weighted-theta Fourier-tail and Wiener contraction certificate.
- A02/precision_15_source_transport (*profile replay*).
Fifteen measured source coordinates, Jacobian and covariance transport.
- A02/precision_8x8_workspace (*profile replay*).
Eight-coordinate SMDR output with positive-definite 8x8 covariance.
- A01/qutrit_weyl_27_matrix (*derived exact*).
Sparse 27x27 qutrit-Weyl left-action realization.
- A01/strict_pew_row (*derived exact*).
Promoted P_{EW} source row at the declared one-shared-primitive standard.

Corpus-state cross-checks.

- A01/charged_yukawa_higgs_profile (*profile replay*).
Versioned Yu, Yd, Ye and lambda_H profile packet.
- A14/ckm_prediction_profile (*numeric certified*).
Three selected CKM profile rows and uncertainty comparison.
- A40/neutral_two_primitive_profile (*profile replay*).
Measured two-splitting neutrino profile, masses and Dirac Yukawa rows.

Open boundary (not evidence of closure).

- A05/strict_upgrade_ledger (*open*).
Current 2/9 strict no-knob upgrade ledger.

No imported row changes theorem ownership or promotes a neighboring claim: all local statements retain their stated hypotheses, domains, and limitations.

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